

2020 AMC 10A Problem 20 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 20

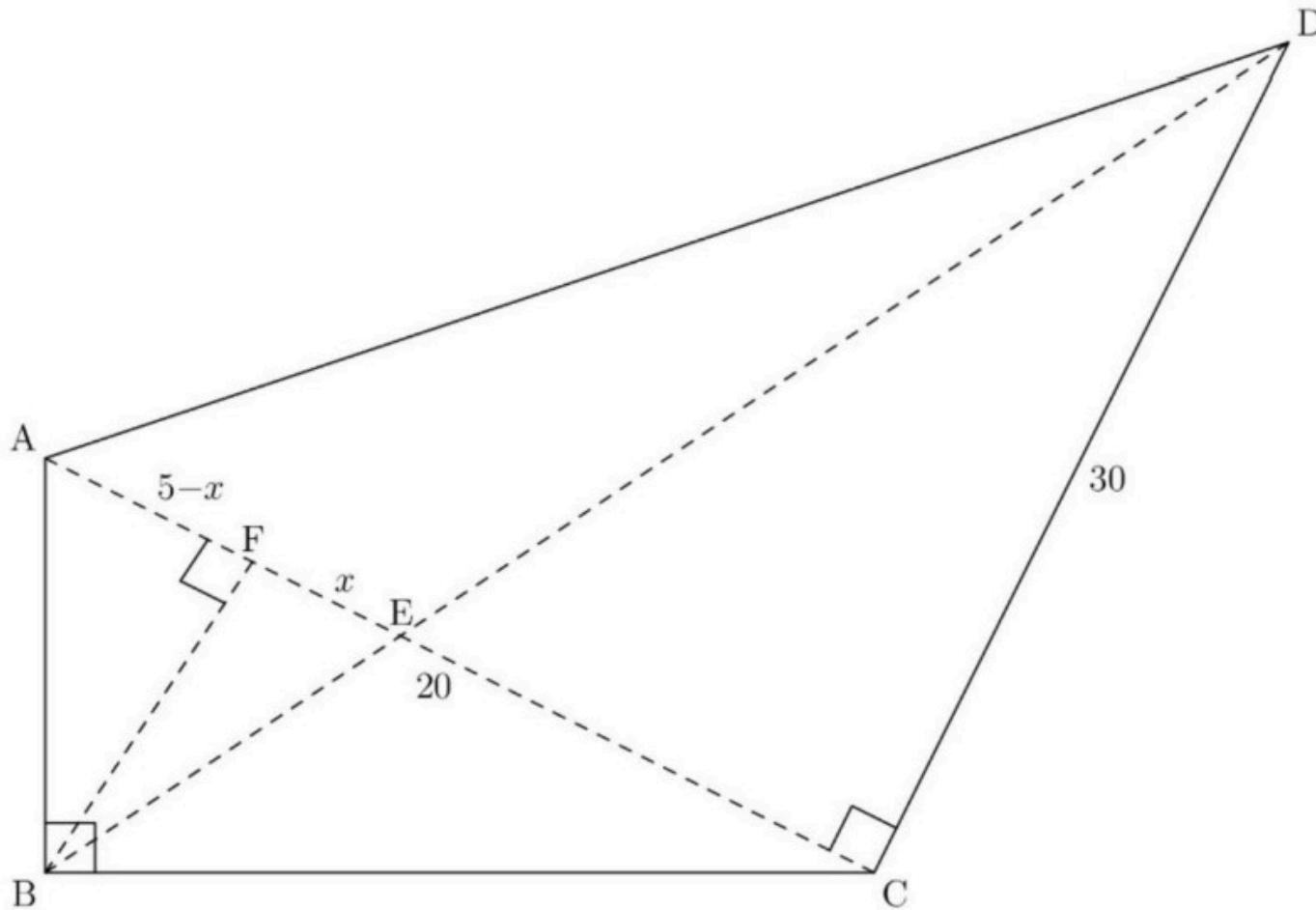
Problem:

Quadrilateral $AECD$ satisfies $\angle ABC = \angle ACD = 90^\circ$, $AC = 20$, and $CD = 30$. Diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point E , and $AE = 5$. What is the area of quadrilateral $ABCD$?

Answer Choices:

- A. 330
- B. 340
- C. 350
- D. 360
- E. 370

Solution:



It's crucial to draw a good diagram for this one.

Since $AC = 20$ and $CD = 30$, we get $[ACD] = 300$.

Now we need to find $[ABC]$ to get the area of the whole quadrilateral. Drop an altitude from B to AC and call the point of intersection F . Let $FE = x$. Since $AE = 5$, then $AF = 5 - x$.

By dropping this altitude, we can also see two similar triangles, $\triangle BFE \sim \triangle DCE$. Since

EC is $20 - 5 = 15$, and $DC = 30$, we get that $BF = 2x$.

Now, if we redraw another diagram just of ABC , we get that

$$(2x)^2 = (5 - x)(15 + x)$$

because of the altitude geometric mean theorem which states that in any right triangle, the altitude squared is equal to the product of the two lengths that it divides the base into.

Expanding, simplifying, and dividing by the GCF, we get

$$x^2 + 2x - 15 = 0.$$

This factors to

$$(x + 5)(x - 3)$$

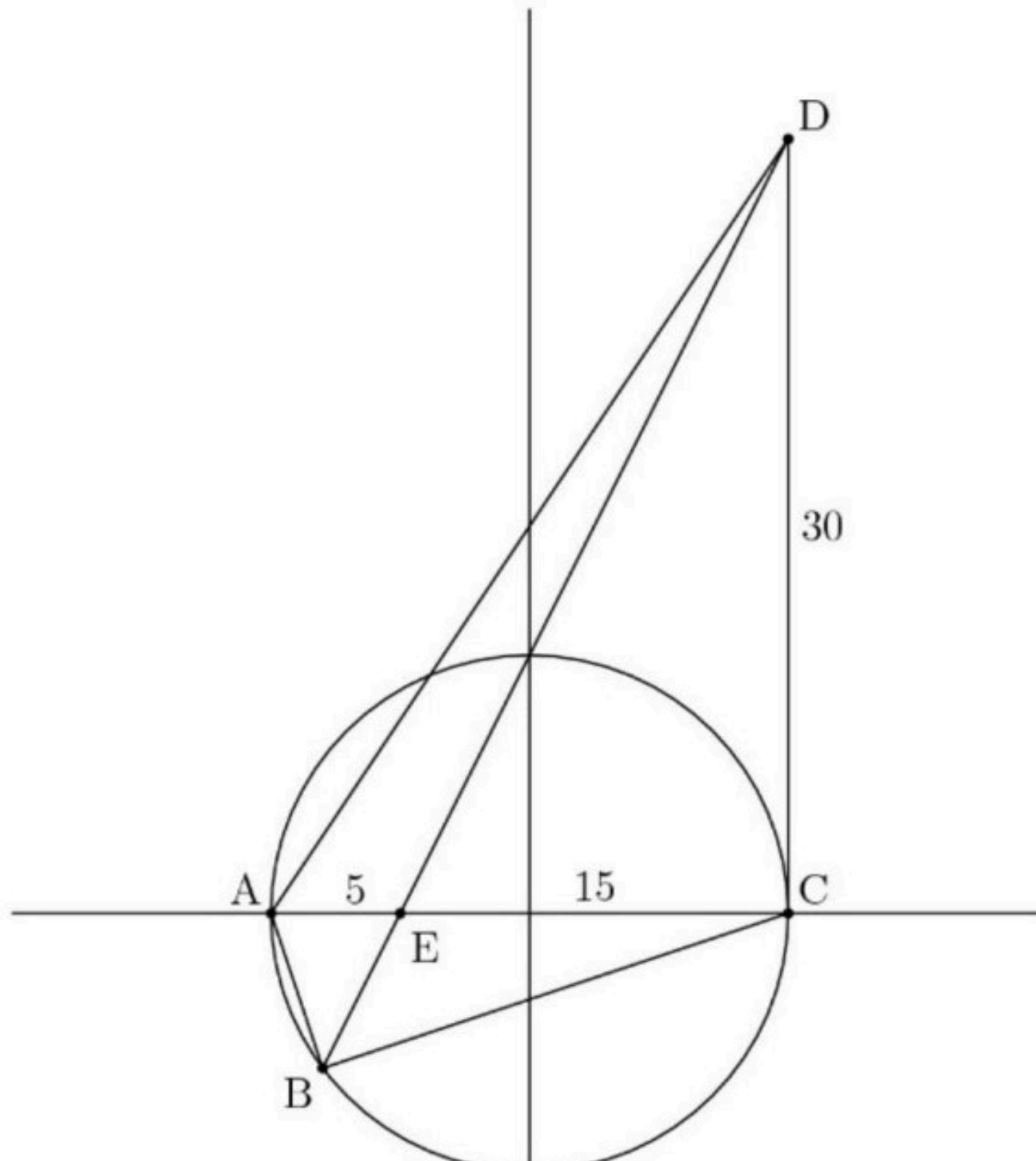
which has roots of

$$x = -5, 3$$

Since lengths cannot be negative, $x = 3$. Since $x = 3$, that means the altitude $BF = 2 \cdot 3 = 6$, or $[ABC] = 60$. Thus

$$[ABCD] = [ACD] + [ABC] = 300 + 60 = \boxed{(D) 360}$$

OR





Let the points be

$$A(-10, 0), B(x, y), C(10, 0), D(10, 30), \text{ and } E(-5, 0)$$

respectively.

Since B lies on line DE , we know that

$$y = 2x + 10$$

Furthermore, since $\angle ABC = 90^\circ$, B lies on the circle with diameter AC , so

$$x^2 + y^2 = 100$$

Solving for x and y with these equations, we get the solutions $(0, 10)$ and $(-8, -6)$. We immediately discard the $(0, 10)$ solution as y should be negative. Thus, we conclude that

$$[ABCD] = [ACD] + [ABC] = \frac{20 \cdot 30}{2} + \frac{20 \cdot 6}{2} = \boxed{(D) 360}$$

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